THEXRA ENTERPRISE SOLUTION PROPOSAL

Architecture Recommendation Brief

Date: 29/09/2026
Status: Draft

1. CLIENT PROBLEM STATEMENT

High online return rates exceeding 25 percent driven by customers misjudging furniture scale and color in their homes prior to purchase.

2. AI RATIONALE & ANALYSIS

"Augmented reality directly eliminates dimensional uncertainty by placing accurate 3D furniture overlays inside customer spaces before checkout. Implementing THEXRA View allows rapid web-based deployment within the required 1-3 month timeline without requiring app store downloads."

3. PROPOSED SOLUTION CONCEPT

Web-based augmented reality product preview enabling shoppers to place true-to-scale 3D furniture models into their physical room using mobile browsers.

4. TECHNOLOGY ARCHITECTURE BLUEPRINT

AR — THEXRA View

1. Overview: THEXRA View embeds a WebXR viewer into the e-commerce storefront. Users activate instant spatial tracking via smartphone cameras to inspect true-to-scale models in real time.

2. User Journey: User views product page → taps View in Room → camera opens → surfaces detected → 3D furniture anchors at 1:1 scale → user verifies fit and clicks buy.

3. Hardware: iOS Mobile Devices, Android Mobile Devices

4. Software: THEXRA View Engine, WebXR Browser Integration

5. AI Components: Light Estimation Machine Learning, Automated Surface Detection ML

6. XR Components: Real-time Spatial Anchoring, True-to-Scale 3D Rendering

7. Backend: Cloud Asset CDN, E-commerce Platform REST API

8. Dashboard: AR Interaction Analytics Dashboard, 3D Asset Management Console

9. Data Flow: User Device → WebXR Camera Feed → Cloud CDN 3D Asset Fetch → Real-Time AR Rendering.

Alternative Option: Mobile/Web — THEXRA View

5. EXPECTED BUSINESS OUTCOMES

Up to 30 percent reduction in product return rates and increased online conversion rates.

6. IMPLEMENTATION NEXT STEPS

Conduct a 3D asset optimization audit and integrate THEXRA View WebXR widget into product detail pages.